

Power Electronics Laboratory

Session 5: Three-Phase DC-AC Converter

Objectives

1. Study the 2-level Three-Phase Voltage Source Inverter (VSI) topology.
2. Analyze switching states, line-to-line voltages, and Common Mode Voltage.
3. Evaluate the impact of Sinusoidal PWM (SPWM) and Overmodulation.
4. Implement Third-Harmonic Injection and Min-Max modulation to extend the linear range.

General instructions

1. All results must be justified with theoretical calculations where applicable.

Part 1: Topology and Switching States

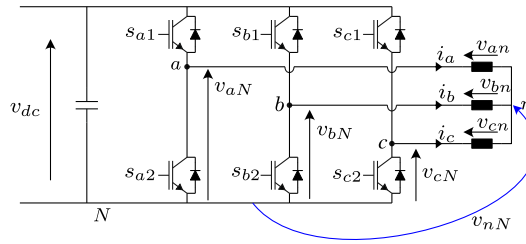


Figure 1: 3-phase 2-level converter.

Table 1: Parameters.

Parameter	Value
V_{dc}	600V
Switching frequency	10 kHz
Three-phase Load	Balanced RL : $L = 15\text{mH}$, $R = 15\Omega$
Load Connection	Star - Floating Neutral

Tasks

1. Complete the following table for the 8 possible switching states of the 2-level converter. Define v_{aN} , v_{bN} , v_{cN} as voltages relative to the negative DC bus, and v_{an} , v_{bn} , v_{cn} as phase-to-load-neutral voltages.
2. Implement the power converter shown in Fig. 1 in PLECS with a proper PWM stage.
3. Choose modulation indexes to generate 360, 300 and 250 V (line-to-line RMS). Plot the converter output voltage relative to the DC negative bus (v_{aN}) and the load phase voltage (v_{AN}). Calculate the magnitude of the load current and the active power for the selected modulation indexes. Verify it in PLECS.
4. **Analysis:** Notice that v_{an} contains a $V_{dc}/2$ component. Explain why this component disappears in the load phase voltage v_{AN} and the line-to-line voltage v_{ab} .

State	S_a	S_b	S_c	V_{an}	V_{bn}	V_{ab}	V_{aN}	V_{nN}
0	0	0	0					
1	1	0	0					
2	1	1	0					
3	0	1	0					
4	0	1	1					
5	0	0	1					
6	1	0	1					
7	1	1	1					

Part 2: Advanced Modulation and Overmodulation

Tasks

1. **Linear vs. Overmodulation:** Gradually increase the modulation index m_x from 0.9 to 1.1 every 0.05. steps. In each case plot 2 cycles the low-frequency component of the output voltage (use a moving average filter tuned to f_{sw}) and its FFT (from 0 to 1kHz). Comment on the appearance of low-frequency harmonics.

2. **Third-Harmonic Injection:** Implement the standard PWM and add a common-mode component:

$$m_{new_a} = M \sin(\omega t) + \frac{M}{6} \sin(3\omega t)$$

Verify that you can increase m_a up to approximately 1.15 (15% increase) without clipping (overmodulation).

3. **Min-Max Injection:** Implement the standard PWM and add a common mode component:

$$m_{new_a} = -\frac{\max(m_a, m_b, m_c) + \min(m_a, m_b, m_c)}{2}$$

Compare the available DC bus utilization with standard PWM.

4. **Common Mode Observation:** For both Third-Harmonic and Min-Max injection, plot the Common Mode Voltage (v_{nN}).
5. **Critical Question:** Even though these techniques inject a heavily distorted signal into the converter legs, the load voltage and current remains perfectly sinusoidal. Justify why.